

Robotic Deployment Preliminary Integrated Milestone Network

Lunar Surface Infrastructure Program

LOGIC INTEGRATION DECISION GATES

Document owner	Program Director
Status	Draft for review
Version	0.1
Date	22 September 2026
Planning basis	Program Charter v1.1; WBS v0.2; Level 4 Work Packages v0.1; RAM v0.1

Purpose: Establish the preliminary dependency logic connecting the 30 Level 4 robotic-deployment work packages, identify parallel workstreams and control gates, and expose logic risks before dates, durations, resources, or commitments are baselined.

Management conclusion: The program has a credible path from fleet readiness through uncrewed surface readiness, but the communications/site-preparation interface requires phased network deployment. This clarification is proposed for controlled incorporation into the WBS dictionary.

1. Director Decision and Intended Use

Approve this version as the logic basis for schedule development. Approval authorizes owners to validate predecessor relationships, define measurable milestone criteria, estimate durations, identify resources, and build an integrated master schedule. It does not establish contractual dates, authorize execution, or confirm a critical path.

What this network answers

- What must finish before another activity can close.
- What may mature in parallel and where progressive planning is appropriate.
- Which decision gates control release of downstream work.
- Where external transportation, cargo, shared-service, and system-owner interfaces constrain the program.
- Which logic chain is most likely to control uncrewed surface readiness once durations are loaded.

What it does not yet answer

- Calendar dates, work-package durations, float, resource feasibility, or cost phasing.
- A confirmed critical path; this requires durations, calendars, constraints, and resource logic.
- Detailed task-level execution sequences within each work package.
- Authorization of crew launch, human-rating certification, procurement, or lunar operations.

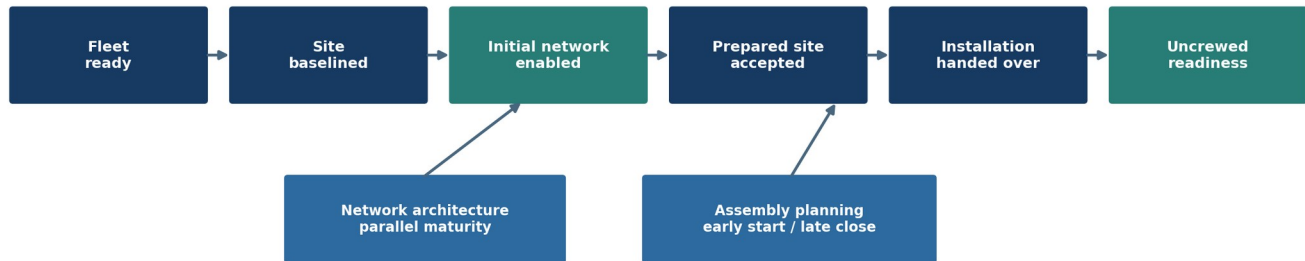
Planning assumptions and logic rules

- Finish-to-start (FS) is the default unless parallel or conditional logic is stated.
- Branch gates require objective acceptance evidence, not only completion claims.
- Progressive work may start on preliminary inputs but cannot close until controlled inputs and receiving-owner concurrence exist.
- External launch and lunar surface delivery are interface milestones owned outside WBS 1.3.
- Changes to controlled predecessor logic require configuration and change-control disposition.
- All models and decisions remain educational and conceptual, not mission-ready engineering.

2. Integrated Milestone Flow

The six Level 3 workstreams converge through evidence-based gates. Communications/navigation and assembly planning mature in parallel; both can become controlling if their acceptance evidence is unavailable when physical surface work or installation is ready to begin.

Program-level milestone logic



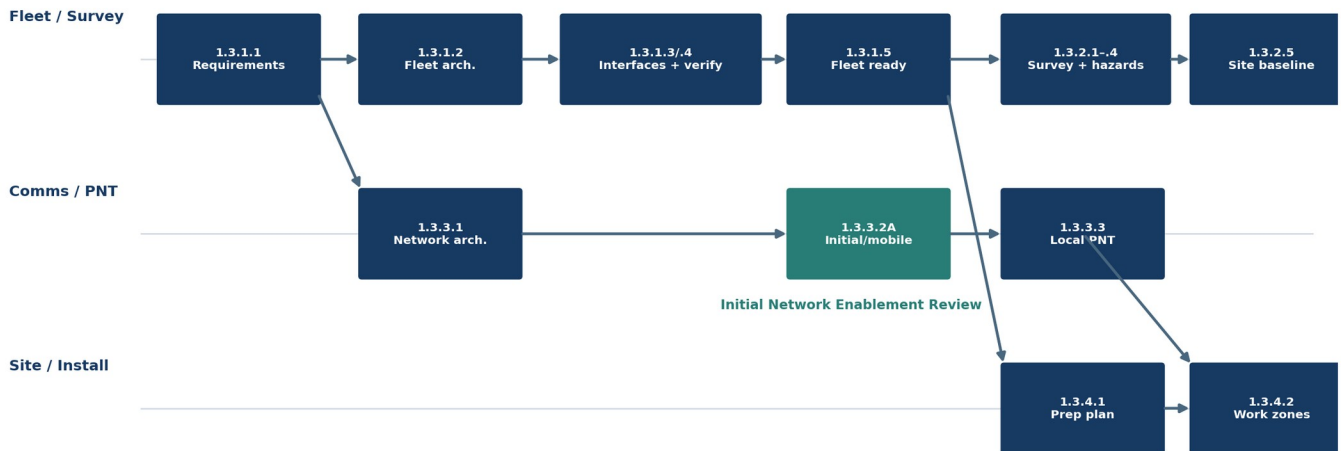
Branch control gates

Branch	Control gate	Closure milestone	Decision evidence
1.3.1	Robotic Fleet Readiness Review	1.3.1.5	Fleet accepted for payload integration
1.3.2	Lunar Site Characterization Review	1.3.2.5	Site baseline approved
1.3.3	Surface Network Readiness Review	1.3.3.5	Final surface network accepted and handed over
1.3.4	Prepared Site Acceptance Review	1.3.4.5	Prepared site accepted and released
1.3.5	Infrastructure Installation Readiness Review	1.3.5.5	Installation baseline handed to system owners
1.3.6	Uncrewed Surface Readiness Review	1.3.6.5	Readiness recommendation dispositioned

3. Upstream Detailed Network

Fleet qualification releases the survey campaign after external surface delivery. Survey evidence establishes the site baseline. In parallel, the network architecture matures and an initial/mobile communications and navigation capability is established before controlled site work begins.

Upstream network: fleet, survey, and initial enablement



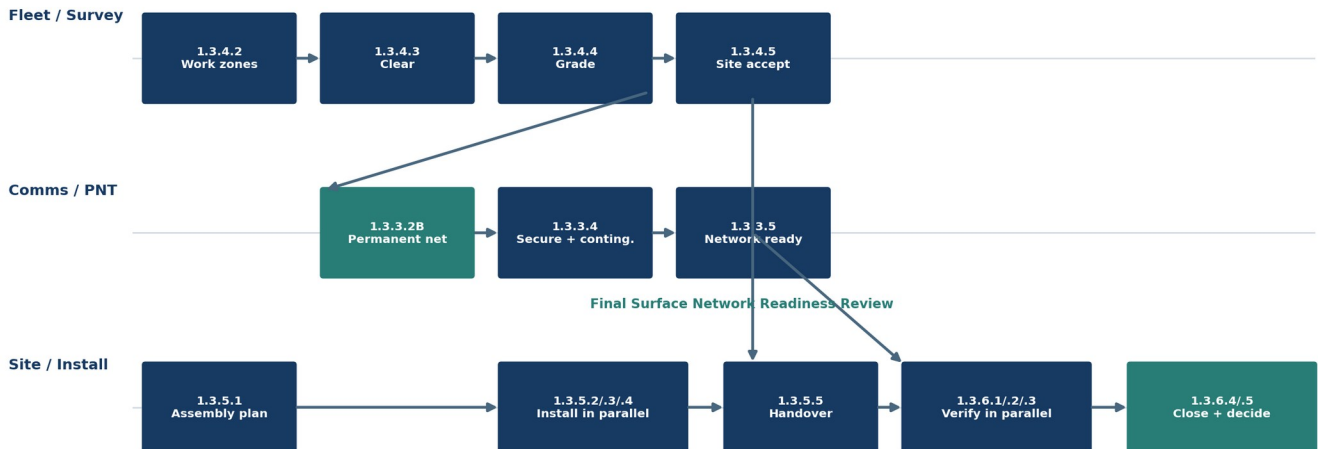
Upstream integration decisions

- The fleet is not released for payload integration until interface, autonomy, fault-management, and qualification evidence converge at 1.3.1.5.
- Terrain/geotechnical and environmental characterization proceed in parallel after survey authorization and converge in the controlled hazard map.
- Site preparation may not enter execution solely on preliminary site data; the site baseline and initial network enablement are entry conditions.
- Network architecture starts before final site selection using candidate-site data, then matures as terrain and hazard evidence is controlled.

4. Downstream Detailed Network

After initial network enablement, work-zone control and site preparation release permanent network locations and accepted construction areas. Installation packages then proceed in parallel where interfaces and cargo permit, before inspection, remote recovery validation, anomaly closure, and the final readiness recommendation.

Downstream network: preparation, installation, and readiness



Downstream integration decisions

- Permanent relay/base-station installation follows limited site preparation while initial/mobile assets preserve command and PNT coverage.
- Assembly planning starts early but cannot close until site, cargo, interfaces, enabling services, and installation constraints are sufficiently mature.
- Cargo offload/staging, utility installation, and equipment placement may overlap only under the approved sequence and simultaneous-operations controls.
- Readiness requires both physical condition evidence and verified remote command, diagnostics, isolation, and recovery capability.

5. Circular Dependency Resolution

Issue: Work package 1.3.3.2 currently depends on prepared locations from WBS 1.3.4, while work-zone control and site-preparation execution depend on communications and local position/navigation/timing capability from WBS 1.3.3. This creates circular logic if both are treated as single finish-to-start packages.

Proposed resolution: Introduce two internal control milestones within 1.3.3.2 without creating new WBS elements:

- 1.3.3.2A — Initial/mobile network deployment: landing assets, mobile relays, and preselected surveyed positions provide the minimum command, telemetry, and PNT capability required to demarcate work zones and begin limited preparation.
- 1.3.3.2B — Permanent network deployment: fixed relays/base stations are installed after limited site preparation produces verified locations, anchoring conditions, and utility access.

Control reviews

- Initial Network Enablement Review — confirms minimum communications, command authority, PNT accuracy, contingency behavior, and coverage before 1.3.4.2 closes.
- Final Surface Network Readiness Review — confirms permanent coverage, performance, cybersecurity, failover, as-built configuration, and operations handover before full downstream readiness closure.

Configuration action: Treat this as a schedule-logic clarification pending approval in the WBS dictionary and change-control record. The work-package owner, network owner, site-preparation owner, safety, systems engineering, and program controls should concur before schedule baseline approval.

6. Candidate Controlling Path

The following chain is the current candidate controlling path. It is a management hypothesis—not a confirmed critical path—because durations, calendars, constraints, resource availability, and total float have not been calculated.

1.3.1.1 → 1.3.1.2 → 1.3.1.3 / 1.3.1.4 → 1.3.1.5 → external launch and surface delivery → 1.3.2.1 → 1.3.2.2 / 1.3.2.3 → 1.3.2.4 → 1.3.2.5 → 1.3.4.1 → 1.3.4.2 → 1.3.4.3 → 1.3.4.4 → 1.3.4.5 → 1.3.5.2 / 1.3.5.3 / 1.3.5.4 → 1.3.5.5 → 1.3.6.1 / 1.3.6.2 / 1.3.6.3 → 1.3.6.4 → 1.3.6.5

Potential alternate controlling paths

- Surface communications/navigation: architecture, deployment, PNT, cyber/contingency integration, or final verification may control site work and remote readiness.
- Cargo and transportation: launch integration, surface delivery, cargo condition, or staging access may control installation.
- Infrastructure interfaces: late interface-control data may hold assembly planning, utility pathways, placement, and turnover.
- Maintenance and spares: long-lead parts or an unproven robotic repair method may hold the uncrewed readiness decision.

Parallel workstream summary

- Terrain and environmental characterization proceed together after survey authorization.
- Network architecture and preliminary assembly planning begin while fleet/site evidence matures.
- Utility installation and equipment placement may execute in parallel after site acceptance and sequence authorization.
- Condition baseline, maintenance readiness, and remote recovery validation converge before anomaly closure.

7. External Interfaces and Schedule-Logic Risks

External interfaces and constraints

- Element 1.2 transportation: payload integration, launch, lunar delivery, surface release, landing constraints, and cargo documentation.
- Habitat, data-center, and shared-infrastructure owners: layouts, interface-control data, acceptance criteria, commissioning prerequisites, and receiving-owner concurrence.
- Common power and external relay services: availability, connection authority, capacity, timing, and degraded-mode support.
- Program governance: safety disposition, configuration control, risk acceptance, budget/resource authorization, and decision-authority availability.

Logic weaknesses requiring validation

Risk / weakness	Required treatment
Progressive design without closure criteria	Early network and assembly planning could remain indefinitely preliminary. Define maturity thresholds and controlled inputs.
External delivery as a single event	Launch, transit, landing, release, inspection, and staging should become explicit interface milestones in the integrated schedule.
Hidden resource conflicts	The same robotic assets may be assigned to survey, network deployment, site preparation, cargo handling, and inspection. Resource loading is required.
Overlapping surface work	Parallel installation can create collision, dust, power, communications, and exclusion-zone conflicts. Tie overlap to approved simultaneous-operations controls.
Acceptance evidence latency	Data processing, discrepancy closure, and decision-board availability may control gates even after physical work ends.
Maintenance readiness closure	Spares and repair planning may begin early, but cannot close until configuration, condition, reliability, and access evidence are known.

8. Milestone Dictionary

This dictionary records the current primary logic and closure evidence for all 30 Level 4 work packages. Detailed internal tasks, supporting milestones, duration estimates, and resource assignments will be developed within the integrated master schedule.

1.3.1 Robotic Fleet and Payload Configuration

Branch gate: Robotic Fleet Readiness Review. A qualified robotic fleet and payload configuration capable of executing the approved survey, preparation, installation, inspection, and contingency missions.

WBS	Milestone / work package	Primary predecessor(s)	Logic	Exit evidence
1.3.1.1	Robotic Mission Concept of Operations and Requirements	Program Charter v1.1; WBS v0.2	FS	Requirements baseline approved
1.3.1.2	Fleet Architecture and Role Allocation	1.3.1.1	FS	Fleet architecture approved
1.3.1.3	Payload Tooling and Interface Configuration	1.3.1.2	FS	Payload/tool interfaces controlled
1.3.1.4	Mobility Autonomy and Fault Management Verification	1.3.1.1; 1.3.1.2; 1.3.1.3	FS / parallel verification	Autonomy and fault management verified
1.3.1.5	Integrated Fleet Qualification and Launch Readiness	1.3.1.3; 1.3.1.4	FS	Fleet accepted for payload integration

1.3.2 Lunar Site Survey and Hazard Characterization

Branch gate: Lunar Site Characterization Review. A controlled evidence package characterizing terrain, environment, hazards, and site suitability for landing, traversal, construction, and sustained operations.

WBS	Milestone / work package	Primary predecessor(s)	Logic	Exit evidence
1.3.2.1	Survey Data Requirements and Campaign Plan	1.3.1.5; external surface delivery	FS	Survey campaign authorized
1.3.2.2	Terrain Topography and Geotechnical Characterization	1.3.2.1	Parallel	Terrain/geotechnical dataset accepted
1.3.2.3	Illumination Thermal Radiation and Dust Characterization	1.3.2.1	Parallel	Environmental profiles accepted
1.3.2.4	Landing Traverse and Work Zone Hazard Mapping	1.3.2.2; 1.3.2.3	FS	Controlled hazard maps released
1.3.2.5	Site Suitability Recommendation and Baseline	1.3.2.1–1.3.2.4	FS	Site baseline approved

1.3.3 Surface Communications and Navigation Setup

Branch gate: Surface Network Readiness Review. An operational surface communications, navigation, and timing capability supporting robotic preparation and later infrastructure operations.

WBS	Milestone / work package	Primary predecessor(s)	Logic	Exit evidence
1.3.3.1	Surface Communications and Navigation Architecture	1.3.1.1; preliminary terrain/site data	Parallel / progressive maturity	Surface network architecture approved
1.3.3.2	Relay and Base Station Deployment	1.3.3.1; 1.3.2.4; equipment delivery	Conditional split	Initial/mobile network enabled; permanent nodes later installed
1.3.3.3	Position Navigation and Timing Establishment	1.3.3.2A; survey control points	FS	Local PNT service established
1.3.3.4	Network Integration Cybersecurity and Contingency	1.3.3.1; 1.3.3.2A; 1.3.3.3	FS / parallel integration	Secure command and contingency paths accepted
1.3.3.5	Coverage Performance Verification and Handover	1.3.3.2B; 1.3.3.3; 1.3.3.4	FS	Final surface network accepted and handed over

1.3.4 Site Preparation and Grading

Branch gate: Prepared Site Acceptance Review. Prepared, stabilized, and verified surface areas and routes suitable for infrastructure installation and robotic operations.

WBS	Milestone / work package	Primary predecessor(s)	Logic	Exit evidence
1.3.4.1	Site Preparation Plan and Equipment Readiness	1.3.1.5; 1.3.2.5; initial network enablement	FS	Site-preparation entry criteria approved
1.3.4.2	Work Zone Demarcation and Hazard Controls	1.3.4.1; 1.3.3.3	FS	Work zones and hazard controls accepted
1.3.4.3	Obstacle Removal and Surface Clearing	1.3.4.2	FS	Work areas and routes cleared
1.3.4.4	Grading Compaction and Foundation Preparation	1.3.4.3	FS	Pads, berms, trenches, and foundations verified
1.3.4.5	Prepared Site Acceptance Survey and Handover	1.3.4.2–1.3.4.4	FS	Prepared site accepted and released

1.3.5 Infrastructure Deployment and Assembly

Branch gate: Infrastructure Installation Readiness Review. Robotic deployment, placement, assembly, and installation evidence for program infrastructure at accepted lunar surface locations.

WBS	Milestone / work package	Primary predecessor(s)	Logic	Exit evidence
1.3.5.1	Deployment Sequence and Robotic Assembly Planning	Early start: 1.3.1.2; closure: 1.3.2.5, cargo/interface maturity, 1.3.4.5	Progressive / conditional	Integrated deployment sequence approved
1.3.5.2	Cargo Offload Transport and Staging	External surface delivery; 1.3.4.5; 1.3.5.1	FS	Cargo inspected, staged, and released
1.3.5.3	Utility Pathway and Interface Installation	1.3.4.4; 1.3.5.1; delivered hardware	Parallel	Utility pathways and tie points accepted
1.3.5.4	Equipment Placement Anchoring and Assembly	1.3.4.5; 1.3.5.1; 1.3.5.2	Parallel	Equipment placement and assembly accepted
1.3.5.5	Installation Verification and System Handover	1.3.5.2; 1.3.5.3; 1.3.5.4	FS	Installation baseline handed to system owners

1.3.6 Inspection Maintenance and Remote Readiness Verification

Branch gate: Uncrewed Surface Readiness Review. A remotely supportable surface system with established condition baselines, maintenance capability, anomaly closure, and evidence supporting crew-arrival decisions.

WBS	Milestone / work package	Primary predecessor(s)	Logic	Exit evidence
1.3.6.1	Inspection and Condition Baseline	1.3.4.5; 1.3.5.5	FS / parallel inspection	Initial condition baseline accepted
1.3.6.2	Preventive Maintenance Spares and Repair Readiness	Early planning: 1.3.1.2; closure: 1.3.6.1 and configuration evidence	Progressive / conditional	Remote maintenance and spares coverage accepted
1.3.6.3	Remote Command Diagnostics and Recovery Validation	1.3.3.5; 1.3.5.5	Parallel	Remote command, diagnostics, and recovery validated
1.3.6.4	Anomaly Resolution and Configuration Closure	1.3.6.1; 1.3.6.2; 1.3.6.3	FS	Critical anomalies closed; residual items dispositioned
1.3.6.5	Uncrewed Readiness Demonstration and Crew Arrival Recommendation	1.3.6.4; accepted habitat/common-service status	FS	Readiness recommendation dispositioned

9. Baseline Approval and Next Actions

Recommended disposition

- Approve v0.1 for cross-functional logic validation.
- Approve the 1.3.3.2A/1.3.3.2B internal control-milestone concept, subject to WBS dictionary and change-control concurrence.
- Direct each accountable owner to validate predecessors, successors, exit evidence, and external interfaces.
- Convert accepted work-package logic into task-level activities only where execution planning requires it.

Schedule-development sequence

Step	Action	Exit condition
1	Logic validation	Close missing, redundant, circular, or open-ended dependencies.
2	Milestone criteria	Make every gate objectively measurable and assign decision authority.
3	Duration and calendars	Estimate work and review durations; define lunar, ground, test, and decision calendars.
4	Resource loading	Load robotic assets, specialist teams, facilities, test equipment, communications, power, and logistics.
5	Schedule analysis	Calculate critical and near-critical paths, float, constraints, and schedule risk.
6	Baseline decision	Approve the integrated master schedule and establish change-control and status cadence.

Approval record

Role	Name	Disposition	Date
Program Director			
Systems Engineering Lead			
Program Controls Lead			
Robotic Deployment Lead			
Safety / Mission Assurance Lead			

Revision record

Version	Date	Description	Owner
0.1	22 Sep 2026	Initial preliminary integrated milestone network for 30 robotic-deployment work packages.	Program Director

Educational planning artifact — not approved for mission design, flight operations, contracting, or crew-safety decisions.